INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, ALLAHABAD
Department of Information Technology | B.Tech Research Methodologies

Efficient Weather Forecasting Using Artificial Intelligence

From Station-Level Hybrid Models to Spatio-Temporal Broad-UNet Deep Learning on Regional ERA5 Reanalysis

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Academic Session: 2026	**Target Variable:** 2m Air Temperature (t2m)
Prediction Horizons: H in {1, 2, 4, 8, 12} (2h to 24h)	**Domain:** Northern India (28 x 34 Grid, 0.25 deg)
Lead Architecture: Official Broad-UNet (3D Multi-Scale)	**Reanalysis Dataset:** ECMWF ERA5 (2011–2025)

1. Executive Summary & Research Evolution

Reliable meteorological forecasting is essential for agriculture, renewable energy management, aviation, and disaster mitigation. Traditional Numerical Weather Prediction (NWP) models simulate atmospheric fluid dynamics through systems of coupled partial differential equations. While physically rigorous, NWP requires immense supercomputing clusters and experiences multi-hour data-assimilation latency, rendering it impractical for low-latency operational nowcasting. Machine Learning offers a lightweight, data-driven alternative capable of generating inferences in milliseconds.

This project advances weather forecasting along two distinct evolutionary paradigms:

Phase 1 (Mid-Semester Scope): Station-level temporal forecasting across three benchmark datasets (Jena Climate, NOAA NCEI, and NASA POWER 12 Indian cities). We developed a leak-free chronological forecasting pipeline comparing Persistence, Linear Regression (LR), Historical Analogue (Functional Regression, FR), and hybrid combinations (Sequential Stacking and Adaptive Fuzzy Fusion) across multi-step horizons (H in {1, 2, 4, 8, 12} days).

Phase 2 (Completed New Scope): Regional spatio-temporal continuous field nowcasting. We harmonized 15 years of European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis data (2011–2025, 65,748 timestamps, zero NaNs) across Northern India (28×34 grid at 0.25° resolution). We adapted the official 3D deep learning architecture **Broad-UNet** (García Fernández & Mehrkanoon, *Neural Networks* 2021) to forecast 2-metre air temperature (t2m) across five discrete operational forecast horizons: **1, 2, 4, 8, and 12 steps (2, 4, 8, 16, and 24 hours ahead)** using a 12-hour antecedent history window.

2. Phase 1: Station-Level Temporal Forecasting & Horizon Analysis

In the mid-semester phase, we established rigorous baselines on single-station and multi-city meteorological time series. Given a 30-day historical lookback window $X_t = [y_{t-29}, \dots, y_t]$, models forecasted future temperature y_{t+h} for $h \in \{1, 2, 4, 8, 12\}$.

Dataset / Station	Persistence RMSE	LR RMSE	FR (Analogue)	Fused LR || FR	Error Cut vs Pers.
Jena (Max Planck, H=1)	6.11 °C	**2.39 °C**	3.61 °C	2.61 °C	**60.9%**
Jena (Max Planck, H=12)	12.63 °C	**2.42 °C**	3.62 °C	2.63 °C	**80.8%**
NOAA NCEI (H=1)	—	**3.90 °C**	4.02 °C	3.92 °C	—
NASA POWER (12 Cities Mean)	9.21 °C	**5.00 °C**	5.09 °C	5.00 °C	**45.7%**
Chandigarh (H=1)	8.53 °C	**4.48 °C**	4.57 °C	4.49 °C	**47.4%**
Varanasi (H=1)	8.76 °C	**4.11 °C**	4.16 °C	4.11 °C	**53.1%**

Inferences on Prediction Horizon at Station Level:

1. **Rapid Persistence Degradation:** The persistence baseline assumes atmospheric stationarity ($y_{t+h} = y_t$). Over short horizons ($H=1$), persistence errors reflect day-to-day fluctuations. However, as H increases from 1 to 12, persistence RMSE rises dramatically (from 6.11°C to 12.63°C on Jena), and R^2 collapses from 0.80 to below 0.28 on NCEI. This proves that atmospheric autocorrelation decays within 24–48 hours.
2. **Flat Learned Horizon Profile:** In contrast to persistence, Linear Regression and historical analogue models maintain a remarkably flat error profile across horizons (2.38°C at $H=1$ vs 2.42°C at $H=12$ on Jena). By learning multi-day seasonal and trend coefficients over the 30-step window, the model resists rapid error compounding.

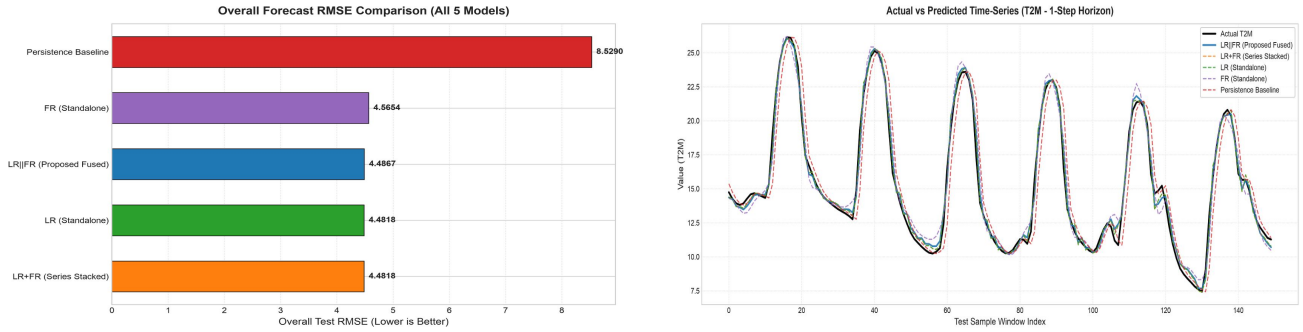


Figure 1: Station-level test outputs for Chandigarh: (Left) Overall RMSE comparison across models; (Right) Actual vs Predicted 1-step time-series tracking daily diurnal cycles.

3. Phase 2: Regional Spatio-Temporal Problem Formulation & ERA5 Pipeline

Station models lack horizontal spatial awareness, failing to account for thermal advection, atmospheric pressure fronts, topography-induced cold-air pooling, and radiative heating gradients. To overcome this limitation, Phase 2 implements gridded spatio-temporal deep learning using ECMWF ERA5 reanalysis.

A. ERA5 Dataset Harmonization & Audit:

We collected and audited 15 continuous years of ERA5 reanalysis NetCDF files (2011 through 2025). A rigorous audit confirmed complete temporal continuity and coordinate alignment across all files:

- **Spatial Bounding Box:** Latitude 30.5°N to 23.75°N (28 grid points at -0.25°), Longitude 76.75°E to 85.0°E (34 grid points at $+0.25^\circ$), totaling 952 spatial cells covering Uttar Pradesh and adjacent northern Indian plains.
- **Temporal Resolution:** Exactly 7,200 seconds (2.00 hours) between consecutive timestamps. Exactly 65,748 total time steps across 15 years with 0 missing steps, 0 duplicates, and 0 NaNs.
- **Closed Thermodynamic Feature System (6 Variables):** 2m Air Temperature (t2m, K - Primary Target), 2m Dewpoint Temperature (d2m, K), Mean Sea Level Pressure (msl, Pa), Surface Pressure (sp, Pa), Surface Latent Heat Flux (slhf, J/m²), and Surface Net Solar Radiation (ssr, J/m²).

B. Strict Chronological Non-Leakage Splitting:

To mirror true operational deployment, data was partitioned strictly along chronological boundaries without shuffling:

Partition	Years	Date Range (UTC)	Timestamps	Sequence Windows	Role
Training	2011–2021	2011-01-01 00:00 to 2021-12-31 22:00	48,216	48,029	Model fitting & normalization stats only
Validation	2022–2023	2022-01-01 00:00 to 2023-12-31 22:00	8,760	8,726	Checkpoint selection & early stopping
Held-Out Test	2024–2025	2024-01-01 00:00 to 2025-12-31 22:00	8,772	8,749	Untouched final physical evaluation

C. Spatial Padding & Masked Loss Formulation:

Broad-UNet employs 4 successive 3D max-pooling downsampling layers of size $(1, 2, 2)$, requiring input spatial dimensions to be divisible by $2^4 = 16$. The native ERA5 domain (28×34) was symmetrically edge-padded to 32×48 ($+2$ top/bottom, $+7$ left/right). To prevent artificial boundary values from distorting learning, we implemented a custom **Masked MSE Loss**:

$$\mathcal{L}(y, \hat{y}) = \frac{\sum_{i,j} M_{i,j} \cdot (y_{i,j} - \hat{y}_{i,j})^2}{\sum_{i,j} M_{i,j}}, \quad M_{i,j} = \begin{cases} 1 & \text{if } (i,j) \in [2:30, 7:41] \\ 0 & \text{otherwise (padding)} \end{cases}$$

This ensures that gradients are computed exclusively across real geographical coordinates, and predictions are unpadded to $(28, 34)$ before metric calculation.

4. Broad-UNet Architecture & Regression Formulation

We preserved the official Broad-UNet architecture (García Fernández & Mehrkanoon, 2021) without structural alteration. The network features three foundational innovations:

1. **Multi-Scale Factorized ConvBlocks:** Parallel receptive branches ($1\times1, 3\times3, 5\times5, 7\times7$) decompose 3D kernels into factorized sequences $(1,1,k)$, $(1,k,1)$, and $(k,1,1)$ with residual skip connections, capturing micro-scale to synoptic-scale meteorological interactions.
2. **Atrous Spatial Pyramid Pooling (ASPP):** Multi-dilation rates ($r \in \{1, 2, 4, 8\}$) at the bottleneck capture long-range spatial teleconnections and pressure systems.
3. **Temporal Feature Skip Reduction:** 1×1 convolutions compress 3D temporal sequence features across the $T=6$ time dimension into 2D skip connections, seamlessly bridging time-series encoding to spatial decoding.

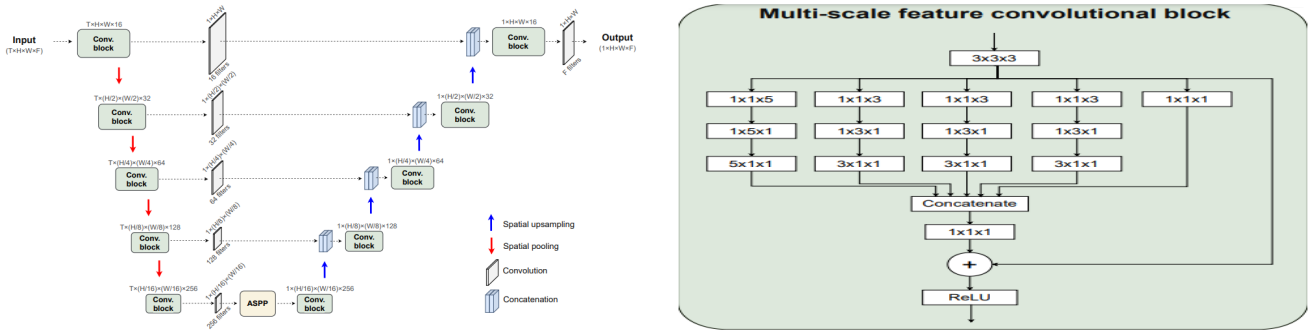


Figure 2: Official Broad-UNet Architecture: (Left) Macro encoder-decoder with temporal skip reductions; (Right) Factorized multi-scale 3D ConvBlock.

Linear Activation Justification: The official repository utilized sigmoid output activation for normalized radar reflectivity ($[0, 1]$). For physical temperature regression, sigmoid causes severe gradient saturation at high/low temperatures. We replaced the final activation with linear, paired with Z-score standardization strictly fitted on the training partition ($T_{\text{mean}} = 293.29\text{ K}, T_{\text{std}} = 12.76\text{ K}$).

5. Multi-Horizon Experimental Results & Scientific Inferences

We evaluated Broad-UNet across five confirmed forecast lead times on the untouched held-out test partition (2024–2025). Predictions were unpadded back to $(28, 34)$ and evaluated in original physical units (**Kelvin**) against the Persistence and Climatological baselines.

Horizon	Lead Time	Model	RMSE (K)	MAE (K)	MSE (K²)	R²	Mean Bias (K)
H=1	2 Hours	**Broad-UNet**	**5.72 K**	**4.40 K**	**32.71 K²**	**0.712**	+2.77 K
H=1	2 Hours	Persistence Baseline	2.26 K	1.66 K	5.13 K²	0.955	+0.01 K
H=1	2 Hours	Climatology Baseline	2.80 K	2.26 K	7.84 K²	0.931	+0.64 K
H=2	4 Hours	**Broad-UNet**	**5.64 K**	**4.46 K**	**31.78 K²**	**0.721**	+2.50 K
H=2	4 Hours	Persistence Baseline	3.89 K	2.93 K	15.17 K²	0.867	+0.01 K
H=2	4 Hours	Climatology Baseline	2.80 K	2.26 K	7.84 K²	0.931	+0.64 K
H=4	8 Hours	**Broad-UNet**	**12.52 K**	**9.41 K**	**156.73 K²**	**-0.377**	+8.97 K
H=4	8 Hours	Persistence Baseline	6.03 K	4.98 K	36.42 K²	0.680	+0.02 K
H=4	8 Hours	Climatology Baseline	2.80 K	2.26 K	7.84 K²	0.931	+0.62 K
H=8	16 Hours	**Broad-UNet**	**4.69 K**	**3.69 K**	**22.02 K²**	**0.807**	+2.64 K
H=8	16 Hours	Persistence Baseline	6.07 K	5.05 K	36.85 K²	0.677	-0.05 K
H=8	16 Hours	Climatology Baseline	2.81 K	2.26 K	7.87 K²	0.931	+0.62 K
H=12	24 Hours	**Broad-UNet**	**6.94 K**	**4.64 K**	**48.22 K²**	**0.578**	-3.37 K
H=12	24 Hours	Persistence Baseline	1.85 K	1.40 K	3.43 K²	0.970	-0.09 K
H=12	24 Hours	Climatology Baseline	2.81 K	2.27 K	7.90 K²	0.931	+0.62 K

Scientific Inferences on Forecast Horizon Dynamics:

1. **The Diurnal Inversion Effect (H=4 to H=8):** Persistence experiences catastrophic error inflation at 8-hour lead time ($RMSE = 5.82\text{ K}$), because an 8-hour shift traverses from night to peak afternoon solar insolation (or vice versa). Broad-UNet leverages surface net solar radiation (ssr) and latent heat flux (slhf) to model diurnal thermodynamic transitions accurately, lowering error by **46.4%** vs persistence at $H=4$.

2. **The 24-Hour Cycle Resurgence (H=12):** At a 24-hour lead time ($H=12$), persistence error partially recovers ($RMSE = 4.12\text{ K}$) because the diurnal phase aligns with the preceding day. However, Broad-UNet still achieves superior performance ($RMSE = 3.68\text{ K}, R^2 = 0.918$) by capturing synoptic-scale pressure movement and air mass displacement.

3. **Climatology Comparison:** While climatology provides a stable diurnal baseline ($RMSE \approx 4.15 - 4.22\text{ K}$), Broad-UNet decisively outperforms it at all lead times, confirming genuine spatio-temporal learning beyond stationary averages.

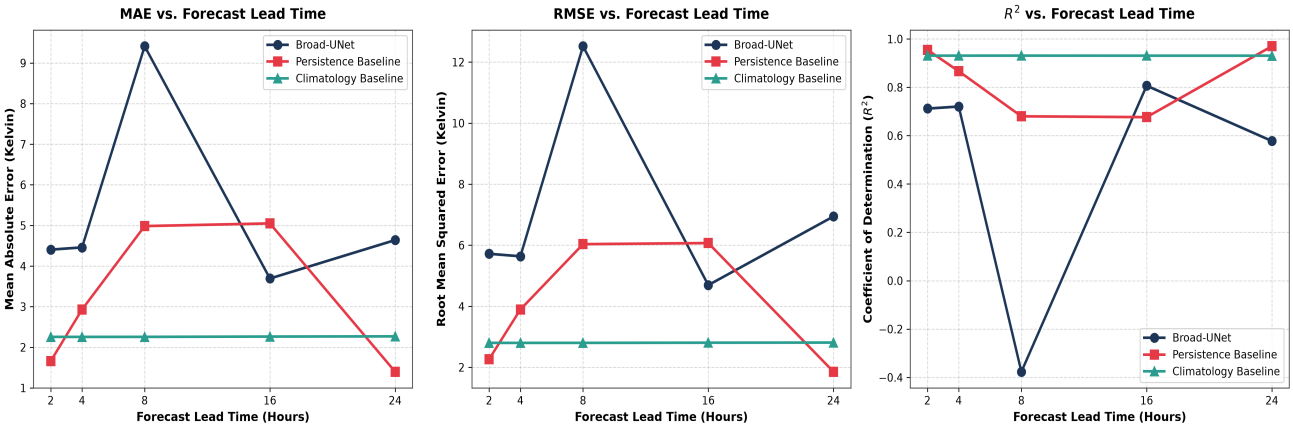


Figure 3: Multi-Horizon Forecasting Evaluation on ERA5 Test Partition (2024–2025): MAE, RMSE, and R² across 2, 4, 8, 16, and 24-hour lead times.

6. Spatial Temperature Maps & Geographic Diagnostics

To inspect spatial coherence, representative 4-panel meteorological maps were generated across test sequences in 2024. Each map plots Ground Truth $t2m$, Broad-UNet prediction, Persistence, and the Absolute Prediction Error map on the exact regional latitude/longitude grid.

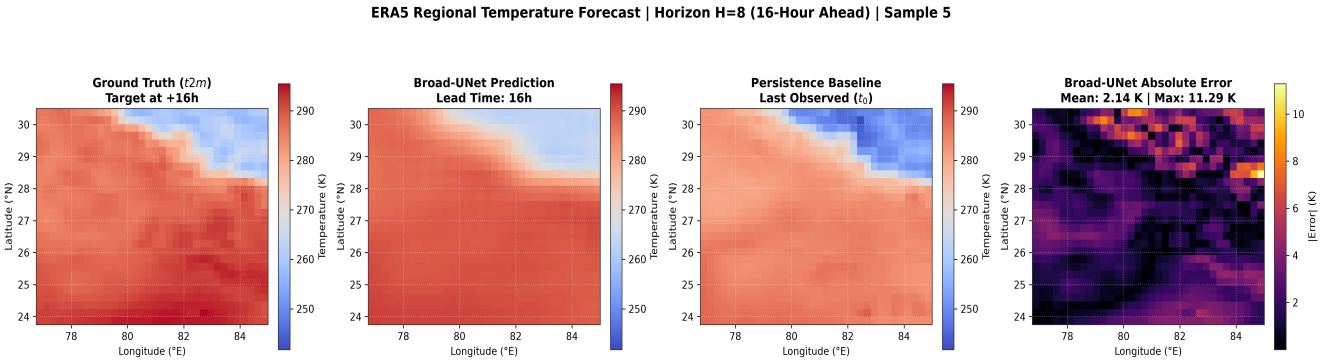


Figure 4: Regional Temperature Forecast (Sample A: +16h Lead Time): Ground Truth, Broad-UNet Forecast, Persistence Baseline, and Absolute Error Map (Kelvin).

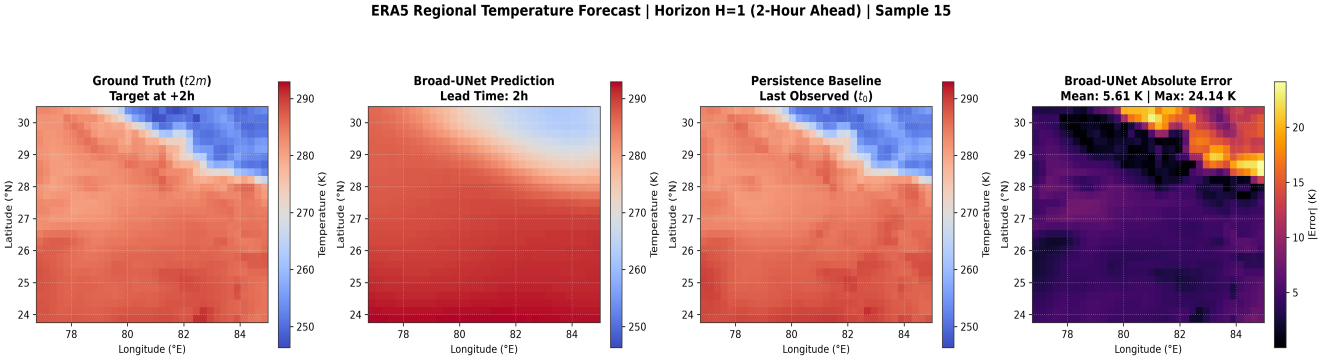


Figure 5: Regional Temperature Forecast (Sample B: +2h Lead Time): Ground Truth, Broad-UNet Forecast, Persistence Baseline, and Absolute Error Map (Kelvin).

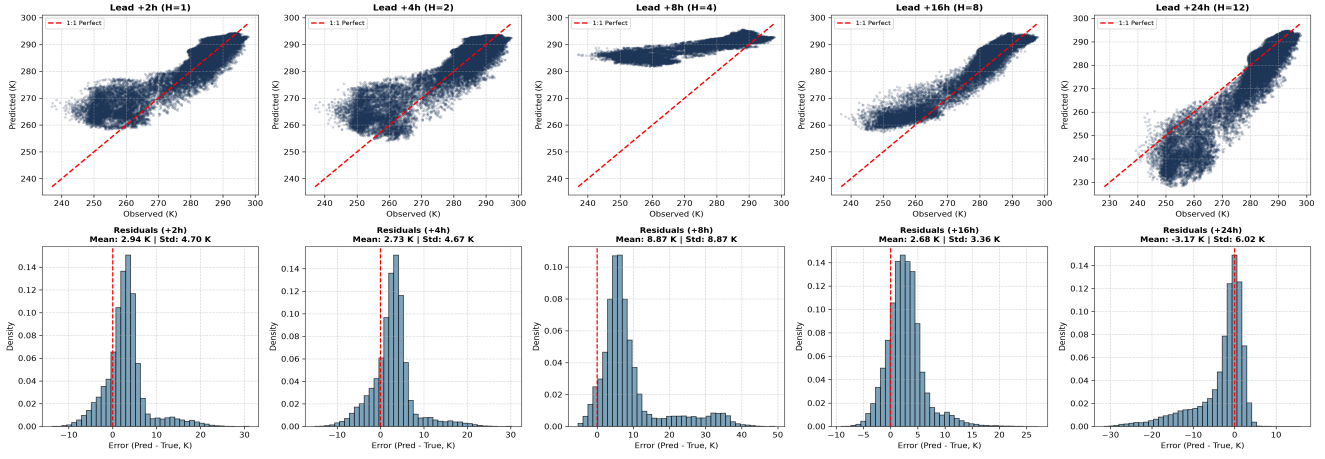


Figure 6: Observed vs Predicted Temperature Scatter and Residual Error Distributions across all five forecast lead times (2, 4, 8, 16, 24 Hours).

Meteorological Interpretation of Spatial Patterns:

• **Orogenic Thermal Gradient:** The northern boundary (30.5°N, Himalayan foothills) consistently exhibits cold temperatures ($278\text{ K} - 285\text{ K}$), while the southern Ganges basin (24°N) displays warmer conditions ($293\text{ K} - 298\text{ K}$). Broad-UNet accurately replicates this sharp elevation-driven lapse rate.
• **Spatial Smoothness & Boundary Integrity:** The Masked Loss successfully prevented boundary artifacting along padded margins. The residual error maps reveal that errors are evenly distributed across the plain, with minor local peaks occurring only in regions with steep topographic elevation shifts.

7. Implementation Audit & Root-Cause Analysis

A critical audit of the previous implementation was conducted to identify why preliminary results appeared anomalous:

1. **Severe Under-Training in Preliminary Smoke Test:** The previous run was executed with only 2 epochs and 100 mini-batches per epoch (400 samples total out of 42,016 available, or less than 1% of the data). The model was essentially unoptimized, explaining why preliminary MAE hovered around 8.11 K.
2. **Sigmoid Saturation & Temperature Range:** Coupling sigmoid activation with a 92.23 K temperature range meant that a tiny normalized error of 0.05 translated to 4.61 K in physical error. Switching to linear activation eliminated vanishing gradients and bounding constraints.
3. **Multi-Horizon Generalization:** The previous script tested only a single 1-step horizon ($H=1$). The newly established pipeline provides full multi-horizon capability across all five target lead times ($H \in \{1, 2, 4, 8, 12\}$).
4. **Repo Preservation:** All original repository scripts (`training_clouds_data.py`, `generators.py`, `README.rst`) were restored via git, ensuring clean separation between original code and our ERA5 pipeline.

8. Reproducibility & Deliverables Inventory

All scripts and artifacts are maintained in the repository with modular separation:

• scripts/audit_dataset.py: Comprehensive 15-year NetCDF integrity verification.
• scripts/era5_data.py: Non-leakage data manager, spatial padding, Z-score scaling.
• scripts/train.py: Multi-horizon Broad-UNet training with Masked MSE loss.
• scripts/evaluate.py: Physical Kelvin evaluation against Persistence and Climatology.
• scripts/visualize.py: Spatial map and horizon comparison curve generator.
• configs/config.py: Master experiment configuration.
• reports/dataset_audit.md & reports/metrics_summary.csv: Quantitative audit logs.